UNIT 6 · STANDARD 6.EE.1

Powers and Exponents

Lesson 6-1

Name: ____________________ Date: ____________________ Class: ____________________

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$$5^{3}$$

$5 \times 5 \times 5$

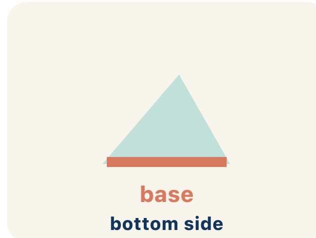
In 2^3, the small 3 means multiply 2 by itself 3 times: $2 \times 2 \times 2 = 8$

Exponent

Write the definition:

____

____



In 5^2, the base is 5 — it is the number being multiplied: $5 \times 5 = 25$

Base

Write the definition:

____

____

$$5^{3}$$

$5 \times 5 \times 5$

$10^3 = 10 \times 10 \times 10 = 1{,}000$ — read as '10 to the third power' or '10 cubed'

Power

Write the definition:

____

____

$$3x + 5$$

no equal sign

Evaluate 3^4: write $3 \times 3 \times 3 \times 3$, then multiply step by step: $9 \times 9 = 81$

Evaluate

Write the definition:

____

____

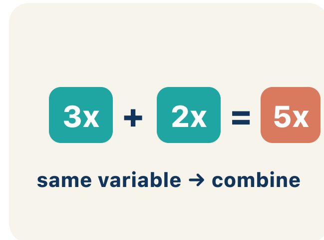
In $3x$, the 3 is the coefficient — if $x = 4$, then $3x = 3 \times 4 = 12$

Coefficient

Write the definition:

___

___



$4x$ and $2x$ are like terms (both have x); $4x$ and $4x^2$ are NOT like terms (different powers)

Like terms

Write the definition:

___

___

Guided Notes **Level 2 Standard**

🎯 **WHAT WE'RE LEARNING TODAY**
I can write and evaluate numbers in exponent form using a base and a power.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Exponent | **Base** | **Power** | **Evaluate** | **Coefficient** | **Like terms**

1. The small raised number that shows how many times the base is multiplied is the ____________ .

2. The number being multiplied repeatedly in a power, like the 3 in 3^4, is the ____________ .

3. A number written with a base and an exponent, like 3^4, is a ____________ .

4. To find the value of an expression, like $2^3 = 8$, is to ____________ it.

5. A number multiplied by a variable, like the 5 in $5x$, is a ____________ .

6. Terms with the same variable raised to the same power, like $3x$ and $7x$, are ____________ .

🧮 Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 I do — watch

Follow each step as your teacher solves it.

Problem: What is the value of 4^3?

A. 64

B. 12

C. 43

D. 7

Step 1 $4^3 = 4 \times 4 \times 4 = 64$.

✅ **Answer:** A. 64

🤝 We do — together

Solve this one with your class using the same steps.

Problem: What is the value of 6^2?

A. 36

B. 12

C. 62

D. 8

Step 1 ____________________

Step 2 ____________________

Answer: ____________________

✏️ You do — your turn

Now try one on your own. Show every step.

Problem: Which expression shows 5^3 as repeated multiplication?

A. $5 \times 5 \times 5$

B. 5×3

C. $5 + 5 + 5$

D. $3 \times 3 \times 3 \times 3 \times 3$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A synth pattern repeats $7 \times 7 \times 7$. How do you write that as a power?

A. 7^3

B. 3^7

C. 7×3

D. 21^3

Show your work:

2. The bass line doubles for 2^6 counts. What is the value of 2^6?

A. 64

B. 12

C. 32

D. 36

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Sentence starter: After 6 hours there are ____ bacteria because 2^6 = ____. If it grew by adding 2 each hour instead, there would only be ____ bacteria. Exponential growth is faster because ____.

Show your work:

Reflect — Exit Ticket

What is the value of 3^4?

A. 81

B. 12

C. 34

D. 64

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Exponent
2. **Notes 2:** Base
3. **Notes 3:** Power
4. **Notes 4:** Evaluate
5. **Notes 5:** Coefficient
6. **Notes 6:** Like terms
7. **We Try (notes):** A. 36 — *$6^2 = 6 \times 6 = 36$.*
8. **You Try (notes):** A. $5 \times 5 \times 5$ — *5^3 means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.*
9. **Try It 1:** A. 7^3 — *The base 7 is used as a factor 3 times, so the power is 7^3. $7^3 = 7 \times 7 \times 7 = 343$.*
10. **Try It 2:** A. 64 — *$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$. The exponent 6 counts how many times 2 is a factor.*
11. **Exit Ticket:** A. 81 — *$3^4 = 3 \times 3 \times 3 \times 3 = 81$.*